L4: Lesson Summary

The goal of this module was to introduce you to real networks, their degree distributions and the differences they have with random networks. From there we explored the Power-law degree distribution, how to test for it, methods of plotting it, and the special properties of networks that have this distribution. These properties include the maximum degree, the robustness of these networks reason that power-law network are sometimes also referred to as scale-free networks. Additionally, we introduced you to several models for generating real networks such as the configuration model, the link-selection model and the preferential attachment model.

Because real networks with power-law degree distributions are the ones that occur in nature, we will continue to build upon these concepts further in the remainder of the modules as we explore concepts such as sociological networks, communities within networks, and the dynamics of contagion spreading withing a network.